

1. Flow Regulation

Recovery Data Curation

OFFLINE VALIDATION LOSS (MSE ↓)

Policy A (w/ Recov)



Policy B (Sanitized)



▼ RANKING INVERSION ON HARDWARE ▼

CLOSED-LOOP SURVIVAL (130 kPa Drop)

Policy A (w/ Recov)



Policy B (Sanitized)



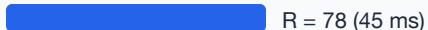
Pruning recoveries hides compounding drift.

2. Surface Touchdown

Penalty Solver Exploit

SIMULATED EPISODE RETURN (↑)

Policy 1 (Braked Descent)



Policy 2 (Unbraked Plunge)



▼ RANKING INVERSION ON HARDWARE ▼

HARDWARE IMPACT FORCE & SURVIVAL

Policy 1 (Braked Descent)



Policy 2 (Unbraked Plunge)



Solver penetration acts as unphysical sink.

3. Regime Migration

Data Debt vs Modeling Debt

SIMULATED / OFFLINE METRIC (↑)

Policy A (BC Imitation)



Policy B (Sim RL)



▼ RANKING INVERSION ON HARDWARE ▼

HARDWARE PRODUCTION COMPLETION

Policy A (BC Imitation)



Policy B (Sim RL)



65 Hz chatter excites resonance and heat.